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Модуль 4

Практика 3

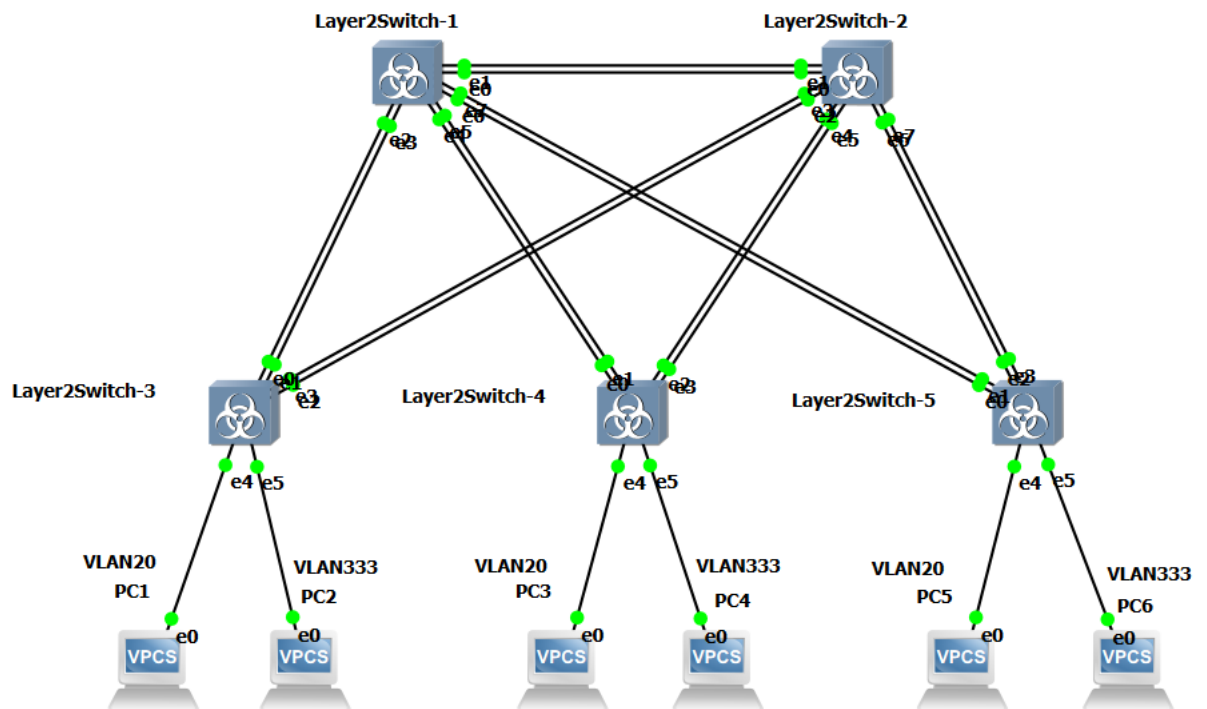


Рисунок 1 – Собранная схема

Для свичей 3 4 5:

enable

configure terminal

interface GigabitEthernet1/0

switchport mode access

switchport access vlan 20

no shutdown

exit

interface GigabitEthernet1/1

switchport mode access

switchport access vlan 333

no shutdown

exit

interface range GigabitEthernet0/0, GigabitEthernet0/2

switchport mode trunk

switchport trunk allowed vlan 20

no shutdown

exit

interface range GigabitEthernet0/1, GigabitEthernet0/3

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 333

no shutdown

exit

Для свичей 1 2:

enable

configure terminal

interface range GigabitEthernet0/0, GigabitEthernet0/2, GigabitEthernet1/0,
GigabitEthernet1/2

switchport mode trunk

switchport trunk allowed vlan 20

no shutdown

exit

interface range GigabitEthernet0/1, GigabitEthernet0/3, GigabitEthernet1/1,
GigabitEthernet1/3

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 333

no shutdown

exit

- Подсеть для VLAN 20: 192.168.20.0 с маской 255.255.255.0
- Подсеть для VLAN 333: 192.168.33.0 с маской 255.255.255.0

PC1 (VLAN 20): 192.168.20.1

PC2 (VLAN 333): 192.168.33.2

PC3 (VLAN 20): 192.168.20.3

PC4 (VLAN 333): 192.168.33.4

PC5 (VLAN 20): 192.168.20.5

PC6 (VLAN 333): 192.168.33.6

ip 192.168.20.1/24

save

ip 192.168.33.2/24

save

ip 192.168.20.3/24

save

ip 192.168.33.4/24

save

ip 192.168.20.5/24

save

ip 192.168.33.6/24

save

Для 3 4 5:

enable

configure terminal

interface range GigabitEthernet0/0, GigabitEthernet0/2

switchport trunk encapsulation dot1q

switchport mode trunk

switchport trunk allowed vlan 20

```
no shutdown
exit
interface range GigabitEthernet0/1, GigabitEthernet0/3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 333
switchport trunk allowed vlan 333
no shutdown
exit
```

Для 1 2:

```
enable
configure terminal
interface range GigabitEthernet0/0, GigabitEthernet0/2, GigabitEthernet1/0,
GigabitEthernet1/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 20
no shutdown
exit
interface range GigabitEthernet0/1, GigabitEthernet0/3, GigabitEthernet1/1,
GigabitEthernet1/3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 333
switchport trunk allowed vlan 333
no shutdown
exit
```

ping 192.168.20.1

ping 192.168.33.2

ping 192.168.20.3

ping 192.168.33.4

ping 192.168.20.5

ping 192.168.33.6

```
PC1> ping 192.168.33.2/24

No gateway found

PC1> ping 192.168.20.3/24

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=15.600 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=8.153 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=10.596 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=13.215 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=8.666 ms

PC1> ping 192.168.33.4/24

No gateway found

PC1> ping 192.168.20.5/24

84 bytes from 192.168.20.5 icmp_seq=1 ttl=64 time=19.787 ms
84 bytes from 192.168.20.5 icmp_seq=2 ttl=64 time=15.481 ms
84 bytes from 192.168.20.5 icmp_seq=3 ttl=64 time=10.491 ms
84 bytes from 192.168.20.5 icmp_seq=4 ttl=64 time=16.605 ms
84 bytes from 192.168.20.5 icmp_seq=5 ttl=64 time=14.422 ms

PC1> ping 192.168.33.6/24

No gateway found

PC1> █
```

Рисунок 2 – Пинги

Для верификации корректности настройки портов доступа, транковых соединений по стандарту IEEE 802.1Q и проверки изоляции виртуальных сетей были проведены тестовые запросы ping с узла PC1 (VLAN 20, IP-адрес 192.168.20.1/24). С другими PC аналогичная ситуация.

ping 192.168.20.3 -c 4

ping 192.168.33.4 -c 4

No.	Time	Source	Destination	Protocol	Length	Info
6	6.690454	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xe44f, seq=1/256, ttl=64 (reply in 7)
7	6.704390	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe44f, seq=1/256, ttl=64 (request in 6)
9	7.705235	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xe44f, seq=2/512, ttl=64 (reply in 10)
10	7.713872	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe44f, seq=2/512, ttl=64 (request in 9)
11	8.715558	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xe44f, seq=3/768, ttl=64 (reply in 12)
12	8.726765	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe44f, seq=3/768, ttl=64 (request in 11)
15	9.730527	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xe44f, seq=4/1024, ttl=64 (reply in 16)
16	9.735711	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe44f, seq=4/1024, ttl=64 (request in 15)
17	10.739748	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xe44f, seq=5/1280, ttl=64 (reply in 18)
18	10.750229	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xe44f, seq=5/1280, ttl=64 (request in 17)

> Frame 6: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
0000 = Priority: Best Effort (default) (0)
....0 = DEI: Ineligible
....0000 0001 0100 = ID: 20
Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.20.3
> Internet Control Message Protocol

Рисунок 3 – Вайлдшарк VLAN 20

No.	Time	Source	Destination	Protocol	Length	Info
10	7.386192	192.168.33.4	192.168.33.2	ICMP	98	Echo (ping) reply id=0x0750, seq=1/256, ttl=64 (request in 9)
12	8.398352	192.168.33.4	192.168.33.2	ICMP	98	Echo (ping) reply id=0x0750, seq=2/512, ttl=64 (request in 11)
16	9.494291	192.168.33.4	192.168.33.2	ICMP	98	Echo (ping) reply id=0x0750, seq=3/768, ttl=64 (request in 15)
18	10.415205	192.168.33.4	192.168.33.2	ICMP	98	Echo (ping) reply id=0x0750, seq=4/1024, ttl=64 (request in 17)
9	7.379094	192.168.33.2	192.168.33.4	ICMP	98	Echo (ping) request id=0x0750, seq=1/256, ttl=64 (reply in 10)
11	8.391544	192.168.33.2	192.168.33.4	ICMP	98	Echo (ping) request id=0x0750, seq=2/512, ttl=64 (reply in 12)
15	9.400549	192.168.33.2	192.168.33.4	ICMP	98	Echo (ping) request id=0x0750, seq=3/768, ttl=64 (reply in 16)
17	10.407544	192.168.33.2	192.168.33.4	ICMP	98	Echo (ping) request id=0x0750, seq=4/1024, ttl=64 (reply in 18)

> Frame 16: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:03 (00:50:79:66:68:03), Dst: Private_66:68:01 (00:50:79:66:68:01)
> Destination: Private_66:68:01 (00:50:79:66:68:01)
> Source: Private_66:68:03 (00:50:79:66:68:03)
Type: IPv4 (0x0800)
[Stream index: 4]
> Internet Protocol Version 4, Src: 192.168.33.4, Dst: 192.168.33.2
> Internet Control Message Protocol

Рисунок 4 – Вайлдшарк VLAN 300

Анализ тегированного трафика (VLAN 20)

На транковом линке между коммутаторами Layer2Switch-3 и Layer2Switch-2 (интерфейс Gi0/2) выполнен перехват ICMP-пакетов при отправке запроса с узла PC1 (192.168.20.1) к PC3 (192.168.20.3). В структуре кадра зафиксировано наличие заголовка стандарта IEEE 802.1Q (802.1Q Virtual LAN), расположенного между заголовками Ethernet II и IPv4.

Значение поля идентификатора сети составляет ID: 20. Наличие этого заголовка подтверждает, что трафик VLAN 20 передается по транковому соединению в тегированном виде.

Анализ нетегированного трафика (VLAN 333 / Native)

На транковом линке для VLAN 333 (интерфейс Gi0/3) выполнен перехват ICMP-пакетов при обмене данными между узлами PC2 (192.168.33.2) и PC4 (192.168.33.4).

В структуре кадра полностью отсутствует заголовок 802.1Q. Поле Type в заголовке Ethernet II указывает напрямую на протокол верхнего уровня IPv4 (0x0800). Отсутствие тега доказывает, что трафик VLAN 333 обрабатывается коммутаторами как Native VLAN на транковом интерфейсе, проходя через линию связи в нетегированном виде.

Добавим Роутер.

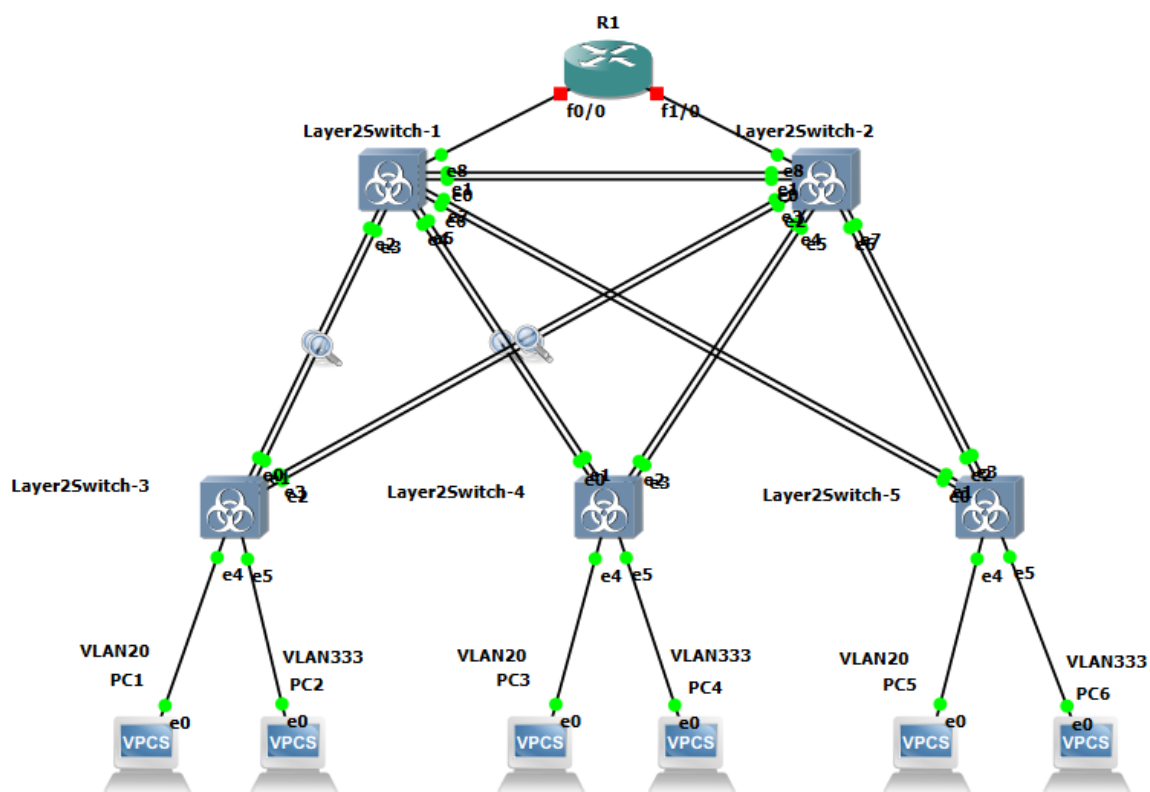


Рисунок 5 – Новая схема

На свиче 1:

enable

configure terminal

interface GigabitEthernet2/0

switchport mode access

switchport access vlan 20

no shutdown

exit

На свиче 2:

enable

configure terminal

interface GigabitEthernet2/0

switchport mode access

switchport access vlan 333

no shutdown

exit

В консоле роутера:

enable

configure terminal

interface FastEthernet0/0

ip address 192.168.20.254 255.255.255.0

no shutdown

exit


```
interface FastEthernet1/0
ip address 192.168.33.254 255.255.255.0
no shutdown
exit
```

Чтобы компьютеры понимали, что для отправки пакетов в «чужую» сеть нужно обращаться к роутеру, настроим на них IP-адреса вместе с адресом шлюза (GateWay).

```
clear ip
ip 192.168.20.1/24 192.168.20.254
save
clear ip
ip 192.168.20.3/24 192.168.20.254
save
clear ip
ip 192.168.20.5/24 192.168.20.254
save
clear ip
ip 192.168.33.2/24 192.168.33.254
save
clear ip
ip 192.168.33.4/24 192.168.33.254
save
clear ip
ip 192.168.33.6/24 192.168.33.254
save
```

```
PC1> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=9.321 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=15.042 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=15.041 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=19.738 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=7.689 ms

PC1> ping 192.168.33.4

192.168.33.4 icmp_seq=1 timeout
84 bytes from 192.168.33.4 icmp_seq=2 ttl=63 time=50.502 ms
84 bytes from 192.168.33.4 icmp_seq=3 ttl=63 time=40.475 ms
84 bytes from 192.168.33.4 icmp_seq=4 ttl=63 time=39.055 ms
84 bytes from 192.168.33.4 icmp_seq=5 ttl=63 time=33.565 ms

PC1> ping 192.168.33.4

84 bytes from 192.168.33.4 icmp_seq=1 ttl=63 time=50.876 ms
84 bytes from 192.168.33.4 icmp_seq=2 ttl=63 time=53.581 ms
84 bytes from 192.168.33.4 icmp_seq=3 ttl=63 time=22.956 ms
84 bytes from 192.168.33.4 icmp_seq=4 ttl=63 time=32.825 ms
84 bytes from 192.168.33.4 icmp_seq=5 ttl=63 time=35.307 ms

PC1> ping 192.168.33.6

192.168.33.6 icmp_seq=1 timeout
84 bytes from 192.168.33.6 icmp_seq=2 ttl=63 time=36.674 ms
84 bytes from 192.168.33.6 icmp_seq=3 ttl=63 time=41.121 ms
84 bytes from 192.168.33.6 icmp_seq=4 ttl=63 time=26.770 ms
84 bytes from 192.168.33.6 icmp_seq=5 ttl=63 time=36.949 ms

PC1>
```

Рисунок 6 – Пинги с ПК1

```
PC6> ping 192.168.33.4

84 bytes from 192.168.33.4 icmp_seq=1 ttl=64 time=20.889 ms
84 bytes from 192.168.33.4 icmp_seq=2 ttl=64 time=22.346 ms
84 bytes from 192.168.33.4 icmp_seq=3 ttl=64 time=8.280 ms
84 bytes from 192.168.33.4 icmp_seq=4 ttl=64 time=6.441 ms
84 bytes from 192.168.33.4 icmp_seq=5 ttl=64 time=10.935 ms

PC6> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=63 time=39.019 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=63 time=51.968 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=63 time=34.982 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=63 time=43.178 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=63 time=48.895 ms

PC6> ping 192.168.20.3

192.168.20.3 icmp_seq=1 timeout
84 bytes from 192.168.20.3 icmp_seq=2 ttl=63 time=42.469 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=63 time=53.733 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=63 time=40.531 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=63 time=28.932 ms

PC6> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=63 time=57.324 ms
84 bytes from 192.168.20.3 icmp_seq=2 ttl=63 time=42.271 ms
84 bytes from 192.168.20.3 icmp_seq=3 ttl=63 time=30.803 ms
84 bytes from 192.168.20.3 icmp_seq=4 ttl=63 time=47.306 ms
84 bytes from 192.168.20.3 icmp_seq=5 ttl=63 time=55.321 ms

PC6> 
```

Рисунок 7 – Пинги с ПК6

Запросы ping с узла PC1 (VLAN 20) до удаленных узлов PC4 и PC6 (VLAN 333) прошли успешно. Первый пакет в серии отправляется с задержкой (timeout) из-за первоначального процесса разрешения ARP-адресов на интерфейсах роутера. Двусторонняя связность также успешно подтверждена тестами с узла PC6 (VLAN 333) до узлов PC1 и PC3 (VLAN 20).